

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11 – SUB-STRAND 1.3 TAXONOMY II: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: How can we identify and classify living organisms accurately? In this task you will use what you observed and learned in Lessons 1 to 8 to write your own Final Explanation. Answer each prompt in the blank space under it. Use the organisms from our school environment (maize, bean plant, grass, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard, goat) as your evidence. Use the Claim-Evidence-Reasoning (CER) pattern where asked: make a claim, support it with evidence you observed, and give reasoning that links the evidence to the claim. Write scientific terms in English. You may talk through your ideas first in Kiswahili or your mother tongue. At the end, think about how your answers will help your group produce its Local Biodiversity Classification Guide.

Part 1: The problem with our first groupings

In Lesson 1 your group sorted the organisms from around the school (maize, bean plant, grass, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard, goat). (a) Describe how your group sorted them and how other groups' groupings compared with yours. (b) Explain why a standard system of classification is needed so that any person anywhere can identify and study these organisms.

Write your answer here:

Part 2: Classifying plants

Use the maize plant, bean plant, fern and moss to answer. (a) State the observable features that place a plant in Bryophyta, Pteridophyta or Spermatophyta. (b) Use CER to justify the placement of the moss and the fern in different groups. Your evidence must come from what you observed in Lessons 2 and 3.

Write your answer here:

Part 3: Classifying animals

Look at the earthworm, snail, mosquito, butterfly, chicken, lizard and goat. (a) Which observable body features can be used to place these animals into different major phyla? (b) The goat and the chicken are both Chordata. Explain how they can still be separated, and name the class of each. (c) Write a CER statement explaining why the mosquito is placed in Arthropoda.

Write your answer here:

Part 4: Dichotomous keys and viruses

(a) Explain how a dichotomous key works and why different people using the same key get the same answer, while groupings by size, colour or usefulness do not. (b) Write the first two couplets of a dichotomous key that could separate the maize plant, fern and moss, using features you can check by eye or with a hand lens. (c) Viruses are not placed with plants or animals. State their characteristics and give two reasons why they are important in nature or to people in Kenya.

Write your answer here:

Part 5: Final Explanation and your Local Biodiversity Classification Guide

Write a Final Explanation that answers the driving question: How can we identify and classify living organisms accurately? Use the CER pattern. Include (a) a claim, (b) evidence from at least two different lessons and organisms from our school environment, (c) reasoning that links your evidence to the claim, and (d) how your Lesson 1 groupings changed. Then say which features of your group's Local Biodiversity Classification Guide (examples, classification, features used, plant key, animal key, viruses) show this idea.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence from the lessons	Uses specific observations from several lessons (e.g. moss rhizoids, fern sori, earthworm segmentation, key results) and links them to the school organisms.	Uses some observations from the lessons but they are general or from only one or two lessons.	Gives little or no evidence, or evidence that is not from the lessons.
Accuracy of classification ideas	Correctly names and describes divisions, phyla and classes with the features that define them, and explains viruses accurately.	Mostly correct classification with minor gaps or mix-ups in features or terms.	Has major errors or classifies by size, colour or use instead of structural features.
Claim-Evidence-Reasoning	Gives a clear claim, relevant evidence and reasoning that explains why the evidence supports the claim.	Gives a claim and evidence, but the reasoning is brief or incomplete.	Gives a claim without evidence or reasoning, or the parts are not connected.
Dichotomous key and the need for a standard system	Builds correct couplets with contrasting observable features and explains clearly why keys give the same answer for everyone.	Builds a mostly workable key and gives a partial explanation of why keys are reliable.	Key is unclear or not paired, and the need for a standard system is not explained.
Link to the driving question and Local Biodiversity Classification Guide	Directly answers the driving question, shows how thinking changed since Lesson 1, and connects ideas to every	Answers the driving question and connects to some parts of the guide.	Does not clearly answer the driving question or link to the guide.

	part of the guide.		
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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